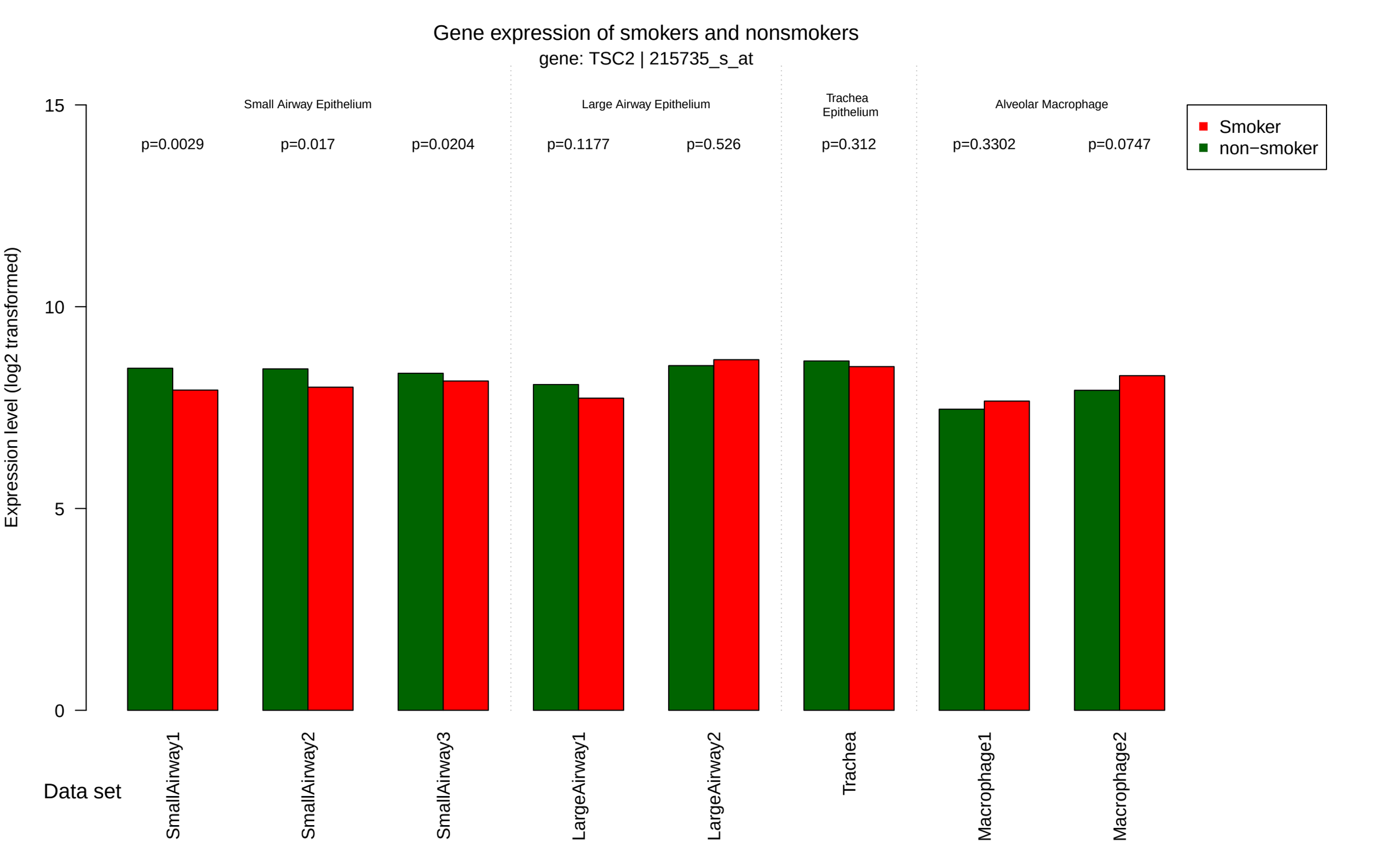
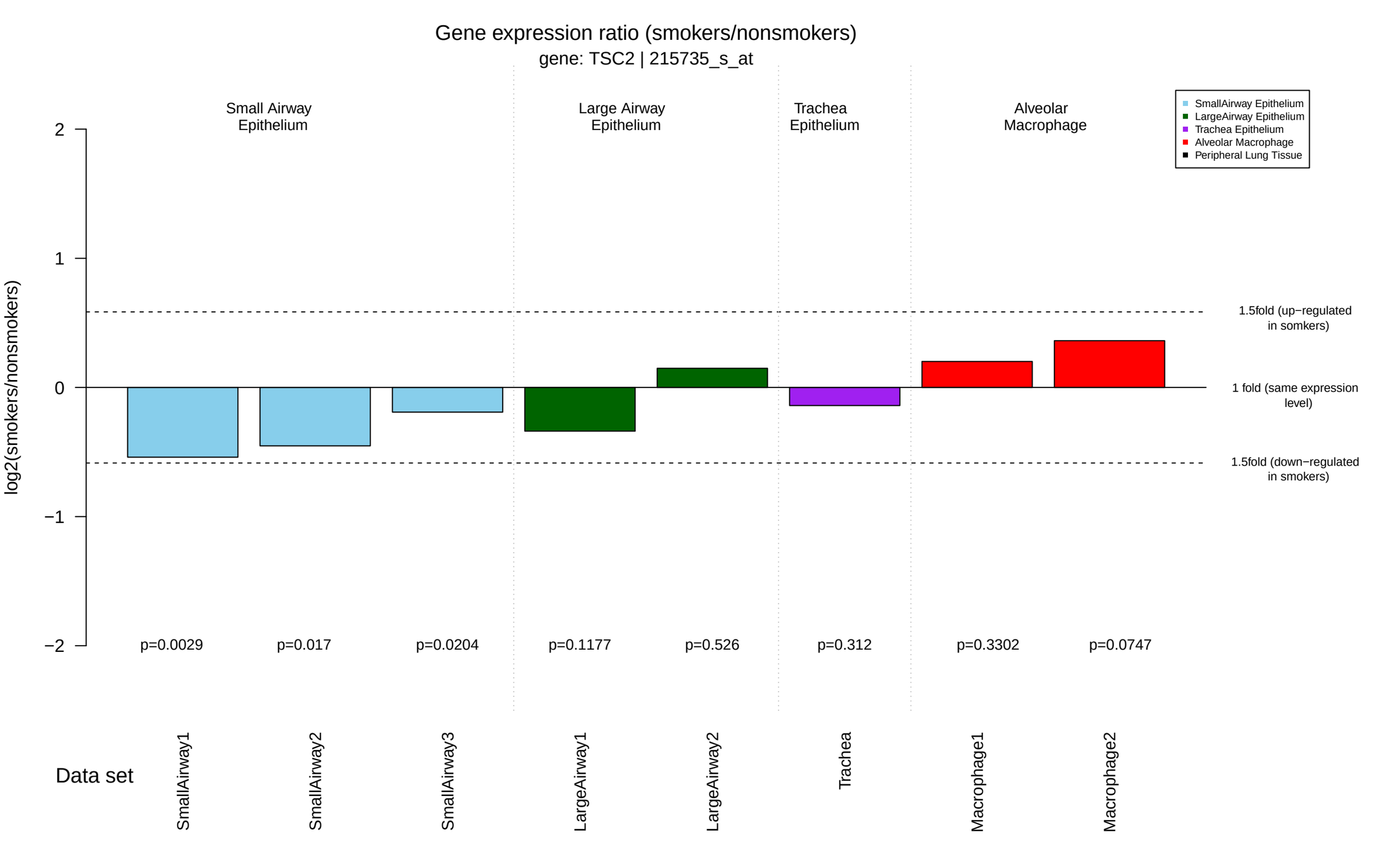
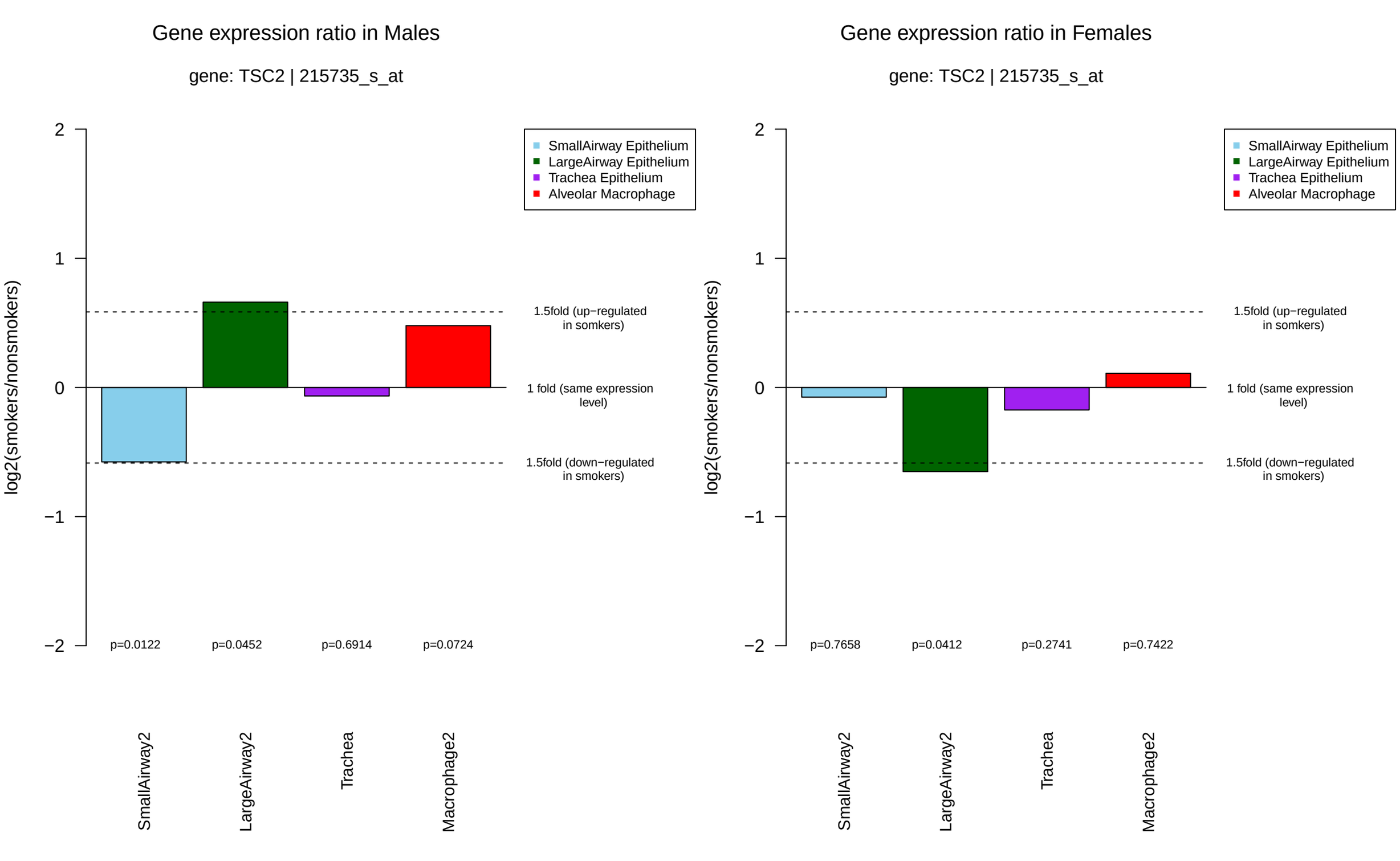
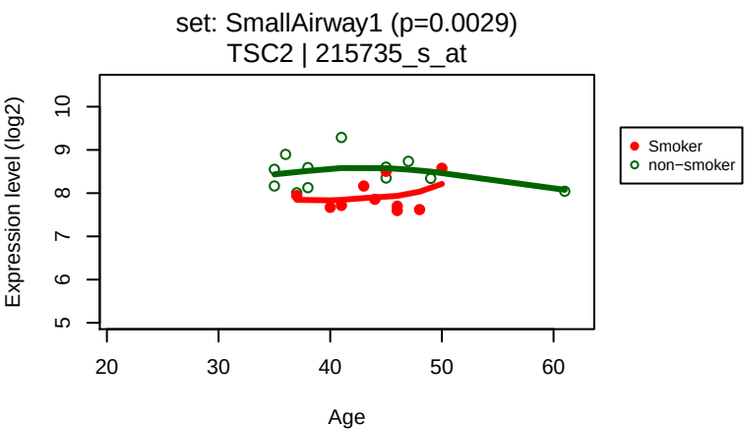
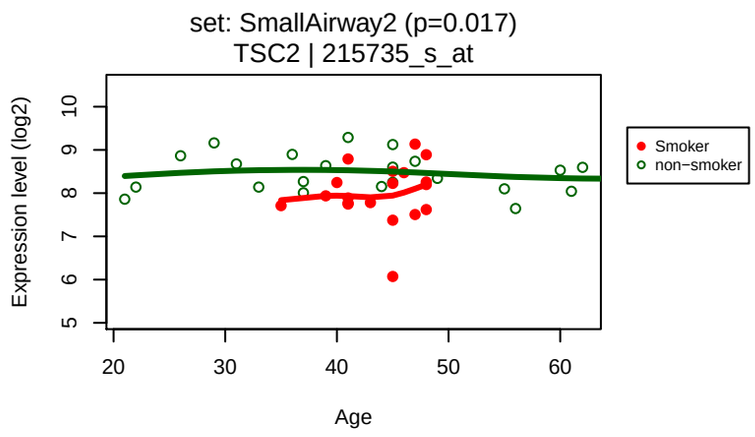
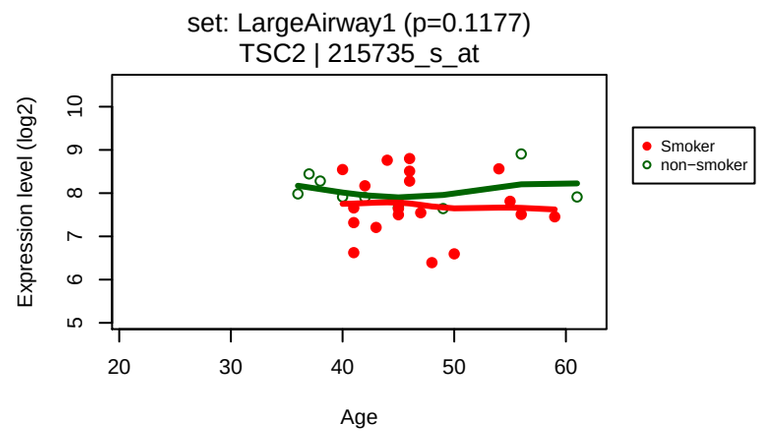

Gene expression of smokers and nonsmokers
gene: TSC2 | 215735_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0029
p=0.017
p=0.0204
p=0.1177
p=0.526
p=0.312
p=0.3302
p=0.0747
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

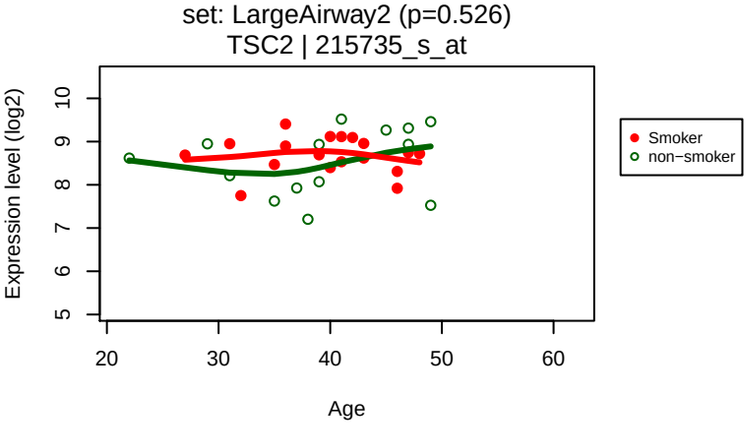

Gene expression ratio (smokers/nonsmokers)
gene: TSC2 | 215735_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0029
p=0.017
p=0.0204
p=0.1177
p=0.526
p=0.312
p=0.3302
p=0.0747
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

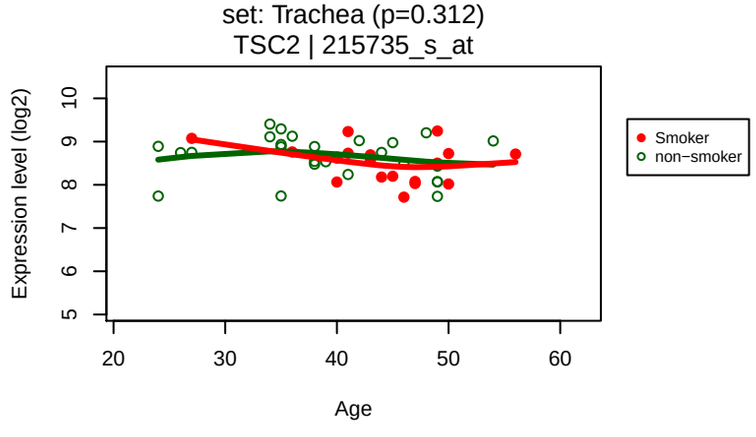

Gene expression ratio in Males
gene: TSC2 | 215735_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.0452
p=0.6914
p=0.0724
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSC2 | 215735_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7658
p=0.0412
p=0.2741
p=0.7422
SmallAirway2
LargeAirway2
Trachea
Macrophage2

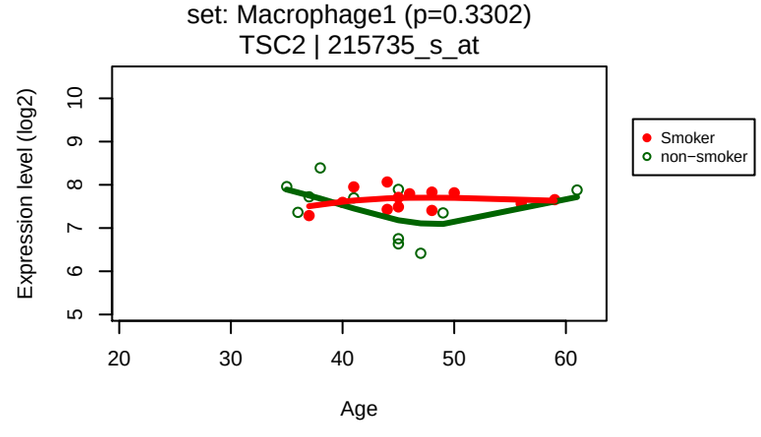

set: SmallAirway1 (p=0.0029)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker

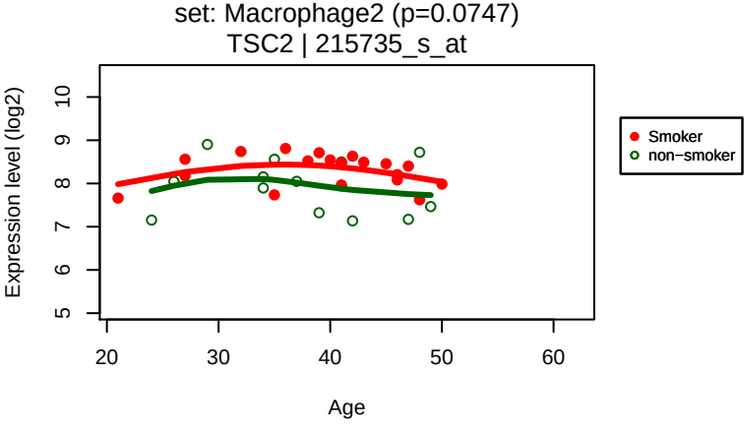

set: SmallAirway2 (p=0.017)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker

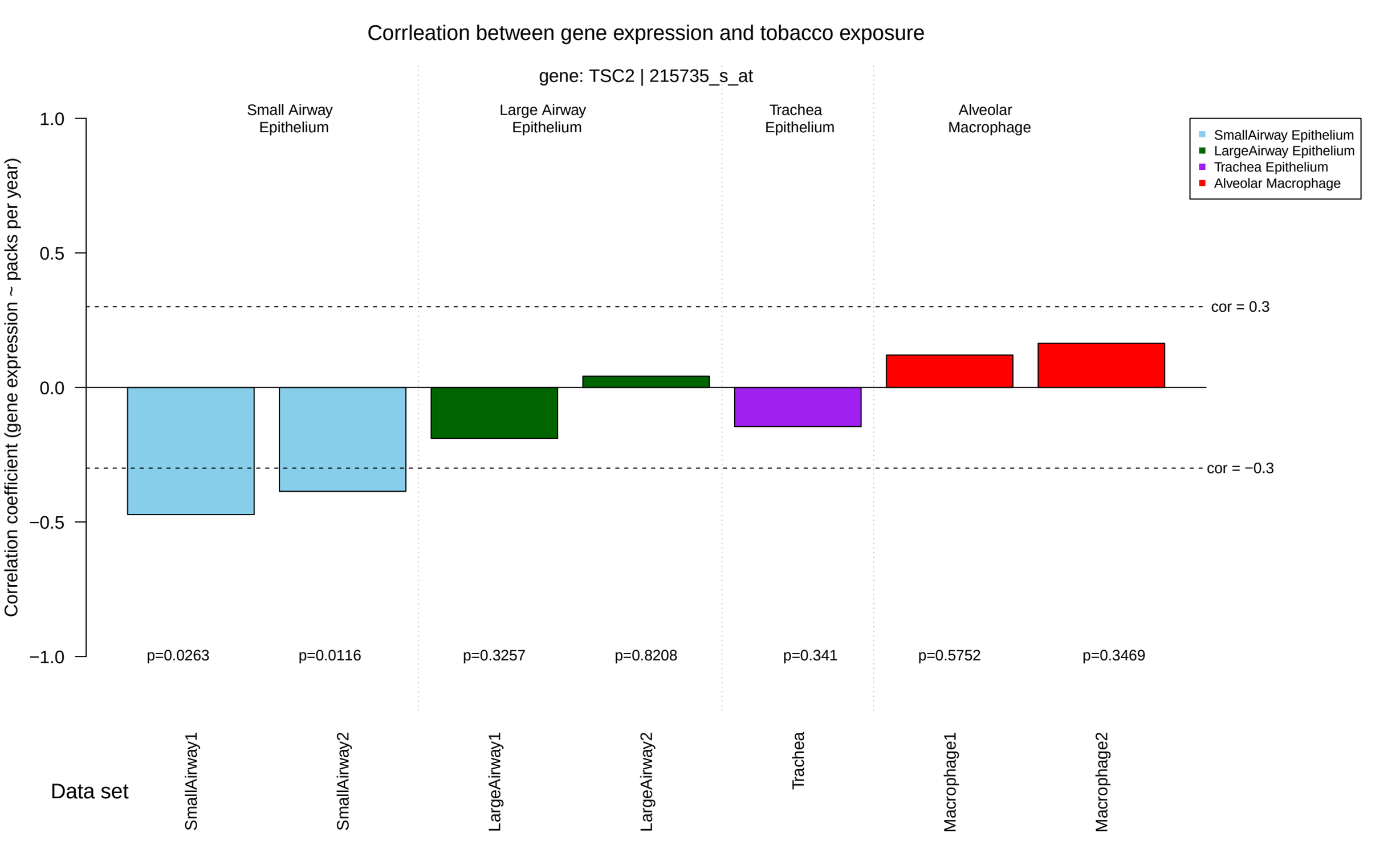

set: LargeAirway1 (p=0.1177)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.526)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.312)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3302)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0747)
TSC2 | 215735_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSC2 | 215735_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0263
p=0.0116
p=0.3257
p=0.8208
p=0.341
p=0.5752
p=0.3469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2